Modeling lunar deformation with gravitational, elastic, and electrostatic interactions using N-body simulation methods

I. Abstract

Context. It is well established that the Moon undergoes deformation under tidal forces. Yet it also carries a small non-neutral charge, a factor often neglected in lunar models. Quantifying such electrostatic effects may be crucial in analyzing the Moon's structural stability.
Aims. We investigate the conditions in which electrostatic, elastic, and gravitational forces substantially influence the Moon's equilibrium geometry. We also aim to determine if internal electrostatic forces contribute to lunar deformation.
Methods. We investigated lunar deformation through developing a model using an N-body approach: the Moon was represented as a 2-dimensional elastic body, composed of N point mass-charges connected by lateral and radial springs. We quantified gravitational forces from the Earth and Sun using time-dependent positional data, elasticity through Hooke's Law, and electrostatics via Coulomb's Law. We then applied the fourth-order Runge-Kutta method (RK4) to predict trajectories and employed nondimensionalization to isolate key parameters.
Results. Our simulations show that electrostatic forces are negligible in lunar modeling, requiring a $\approx 10^{19}$ times higher net charge for perceptible effects. They also reveal the predominance of lateral spring forces over electrostatic, radial spring, and gravitational forces in contributing to deformation, and an unforeseen self-stabilization phenomenon of our system.

Keywords: lunar deformation, planetary modeling, N-body simulation, computational astrophysics

II. Introduction

Analyzing the deformation of planetary bodies is a critical factor in understanding their structural stability, orbital dynamics, and long-term evolution. Previously, models predicting planetary deformation have predominantly focused on tidal interactions and elastic forces. They have also been limited by the fact that they were focused on simple two-body interactions, such as the gravitational relationship between the Earth and the Moon (Kim, 2021). Previous studies suggest a significant impact on results when introducing a third interacting body, such as the Sun, into simulations (Gu et al., 2025).

In this study, we developed a computational model designed to predict the deformation of an arbitrary planet. Contrary to previous models, our formulation allowed us to understand the deformation of planets that contain non-neutral charges. We incorporated these force interactions by approximating electrostatic forces using Coulomb's Law. We also modeled the planet's elasticity with Hooke's Law. In our simplification of our model, we approximated the planet as a collection of N bodies, rather than a continuous distribution of mass. We treated individual planetary components independently, thus functioning as an N-body simulation to capture the three-way interactions of gravitational, electrostatic, and elastic influences. We also introduced real and time-dependent positional data for the Earth and Sun to include their effects on the moon's trajectory.

By applying our model to the moon, our research aims to identify and analyze the parameters that lead to substantial moon deformation. Unlike traditional models that focus exclusively on gravitational and elastic forces, we included electrostatic forces because we found the charge of the moon to be non-negligible. In earlier studies, the average potential difference was determined to be approximately -35 V (Halekas et al., 2002). Using this value and the formula for the capacitance of a sphere, we estimated the Moon's net charge to be -6.776×10^{-3} C, a small value, but with the potential to affect deformation.

Our model aims to explain how combinations of planetary charge, elasticity, and external gravitational fields affect the stability and deformation of the Moon. We explore and analyze the conditions in which instability or significant deformation occurs. The paper is organized as follows. In the Methods section, we describe the construction of our model, including the formulation and force equations employed. In the Results section, we present simulations and the effects of varying key parameters such as net charge and spring stiffness. In the Discussion, we interpret the implications of these findings for lunar evolution and planetary modeling. Finally, we propose directions for future research, including multi-layered structures, 3D modeling, and the effects of other massive bodies such as Jupiter.

III. Methods

Model Formulation

To simulate the deformation under external and internal forces, we constructed a computational model based on physical first principles. First, we simplified our planet into a 2D cross-section, composed of a collection of N point masses uniformly arranged in a unit circle. Each of these points was assigned a mass m/N, where m was the total mass of the moon. Transforming our planet into this geometric configuration allowed us to examine non-rigid body effects, such as stretching and compression, which was crucial for analyzing planet deformation.

Each point also carried a charge q/N, where q was the net charge of the moon. Structural elasticity was modeled using springs, with adjacent points connected with springs of stiffness k_a and each point connected to the geometric center with stiffness k_c. These springs had an adjustable relaxed length of $\mathrm{refdist}$. Adding these adjustable parameters allowed us to study scenarios with different elasticities.

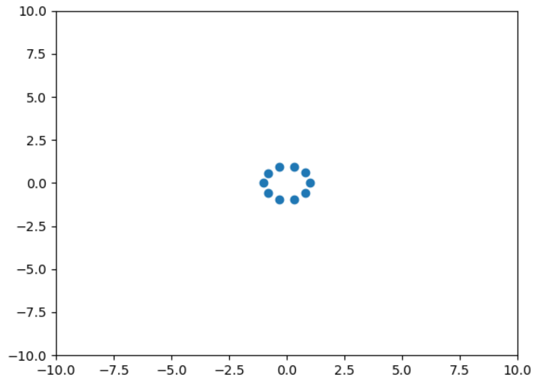

Figure 1. Image from our simulation, taken at time $T = 0$. The moon was modeled as a set of $N = 10$ discrete point masses, with each point having mass $m/10$ and charge $q/10$.

External Bodies: Earth and Sun

We treated the Earth and the Sun as point masses, as their internal deformations were outside the scope of this study. We then obtained their positions using real-time ephemeris data from the Wolfram Data Repository.

We calculated the relative positions of the Earth and the Sun with respect to the center of the planet at each time step. We set the relative positions to $\overrightarrow{r_{12}}(t)$ and $\overrightarrow{r_{13}}(t)$ for the Earth and the Sun, respectively. To scale the planetary system appropriately, all distance values were then normalized by the Moon's average radius $R_0 = 1.7374 \times 10^6$m, which allowed us to set the Moon's radius to 1 in the simulation.

The Earth's mass m_{12} was set to 1, and the Sun's mass $m_{13}$was scaled accordingly to be the ratio of the mass of the Sun to the mass of the Earth: $m_{13} = \frac{1.989\times10^{30}}{5.972\times10^{24}} \approx 3.33 \times 10^5$. Both bodies were assumed to be electrically neutral, so only gravitational forces were calculated between them and the Moon.

Force Model

The net force on body i was calculated by:

$$\overrightarrow{F_\imath} = \sum_{j\neq i} \kappa \frac{q_i q_j}{r_{ij}^3} \cdot \overrightarrow{r_{\imath\jmath}} - \sum_i k_a \overrightarrow{r_{\imath,\imath+1}} - \sum_i k_c \overrightarrow{r_{\imath,11}},$$

where negative terms represent attractive forces and positive terms represent repulsive forces. Dividing the equation by m_i and accounting for the damping term, proportional to velocity to model energy dissipation, gave us the net acceleration equation:

$$\sum_{j\neq i} \kappa \frac{q_i q_j}{r_{ij}^3 m_i} \cdot \overrightarrow{r_{\imath\jmath}} - \sum_i \frac{k_a \overrightarrow{r_{\imath,\imath+1}}}{m_i} - \sum_i \frac{k_c \overrightarrow{r_{\imath,11}}}{m_i} - d\left(v_i \cdot \widehat{r_{\imath,11}}\right)\widehat{r_{\imath,11}},$$

where d was the damping coefficient.

Nondimensionalization

We employed nondimensionalization to increase computational efficiency.

[Include?]
Let x^* denote the nondimensionalized variable x. Let x_0 denote the characteristic value for x. We know that $x = x_0 x^*$. Then, the expression for the gravitational acceleration on body i equates to

$$\sum_{j \neq i} G \frac{m_j}{r^2} = \sum_{j \neq i} \frac{G_0 G^* M_0 m_j^*}{L_0^2 r^{*2}}.$$

We multiply this expression by $\frac{L_0^2}{G_0 M_0}$ for nondimensionalization:

$$\frac{L_0^2}{G_0 M_0} \cdot \frac{G_0 G^* M_0 m_j^*}{L_0^2 r^{*2}} = \frac{G^* m_j^*}{r^{*2}}$$

The expression for the electrostatic acceleration is given by

$$\kappa \frac{q_i q_j}{r^2 m_i} = \frac{\kappa_0 \kappa^* q_0 q_i^* q_0 q_j^*}{L_0^2 r^{*2} M_0 m_j^*}$$

Multiplying by $\frac{L_0^2}{G_0 M_0}$, we get

$$\frac{L_0^2}{G_0 M_0} \cdot \frac{\kappa_0 \kappa^* q_0 q_i^* q_0 q_j^*}{L_0^2 r^{*2} M_0 m_j^*} = \frac{\kappa_0 q_0^2}{G_0 M_0^2} \cdot \frac{\kappa^* q_i^* q_j^*}{r^{*2} m_i^*}.$$

The acceleration from k_a spring forces is given by

$$\frac{k_a r}{m_i} = \frac{k_{a0} k_a^* L_0 r^*}{M_0 m_i^*}.$$

Multiplying by $\frac{L_0^2}{G_0 M_0}$, we get

$$\frac{L_0^2}{G_0 M_0} \cdot \frac{k_{a0} k_a^* L_0 r^*}{M_0 m_i^*} = \frac{k_{a0} L_0^3}{G_0 M_0^2} \cdot \frac{k_a^* r^*}{m_i^*}.$$

Similarly, the acceleration from k_c spring forces is nondimensionalized to be

$$\frac{k_{c0} L_0^3}{G_0 M_0^2} \cdot \frac{k_c^* r^*}{m_i^*}.$$

Thus, the overall nondimensionalized acceleration equation equates to

$$\frac{d^2 r^*}{dt^2} = \frac{G^* m_j^*}{r^{*2}} + \frac{\kappa_0 q_0^2}{G_0 M_0^2} \cdot \frac{\kappa^* q_i^* q_j^*}{r^{*2} m_i^*} + \frac{k_{a0} L_0^3}{G_0 M_0^2} \cdot \frac{k_a^* r^*}{m_i^*} + \frac{k_{c0} L_0^3}{G_0 M_0^2} \cdot \frac{k_c^* r^*}{m_i^*}.$$

We now define the following coefficients:

$$A = \frac{\kappa_0 q_0^2}{G_0 M_0^2}$$
$$B = \frac{k_{a0} L_0^3}{G_0 M_0^2}$$
$$C = \frac{k_{c0} L_0^3}{G_0 M_0^2}.$$

These coefficients govern the relative strength of the electrostatic, lateral spring, and radial spring forces, respectively. For example, if the coefficient A were to be doubled, the electrostatic force would double.

Finding the Characteristic Time

We know that the units of G are $\frac{[L]^3}{[M][T]^2}$. So, we can say $G_0 = \frac{L_0^3}{M_0 T_0^2}$.

Solving for T_0, we get $T_0 = \sqrt{\frac{L_0^3}{G_0 M_0}}$.

Numerical Integration and Simulation Conditions

We employed the fourth-order Runge-Kutta method to approximate the paths of each point. For the conditions, our simulations were conducted using:

- Number of points $N = 10$
- Total time $T = 100$
- Number of timesteps $N_t = 10000$
- Initial angular velocity $\omega_0 = 0.3$
- Damping factor $d = 1$
- Spring relaxed length $refdist = 0$.

IV. Results

We created a function named $simulate$, which accepted three nondimensional coefficients A, B, and C, representing the relative electrostatic, lateral spring, and radial spring forces. This function allowed us to test our system under different force configurations by altering these nondimensionalized coefficients. We also computed the average radius r of the moon over the simulation runtime. Here, r was a dimensionless quantity that represented the moon radius relative to its real-world value. For example, $r = 1$ corresponds to the Moon's actual radius, while $r = 0.5$ would indicate that the simulated body has contracted to half the Moon's actual size.

Calculating the Real-World Values of the Nondimensionalized Coefficients

We determined where the values for A, B, and C approximated the real-world physical values of the moon. First, we calculated A by substituting the following physical constants into our nondimensionalization formula:

$$\kappa_0 = 8.99 \times 10^{9}$$
$$q_0 = 6.776 \times 10^{-3}$$
$$G_0 = 6.67 \times 10^{-11}$$
$$M_0 = 5.972 \times 10^{24}.$$

This yielded in a corresponding value of $A = 1.74 \times 10^{-34}$.

In reality, the Moon does not undergo significant deformation. Thus, we computed the combination of values B and C which preserved the original dynamics, circular structure, and the equilibrium radius of 1. We identified the following coefficient values:

$$A = 1.74 \times 10^{-34}$$
$$B = 2.64 \times 10^{-5}$$
$$C = 1.00 \times 10^{-5}.$$

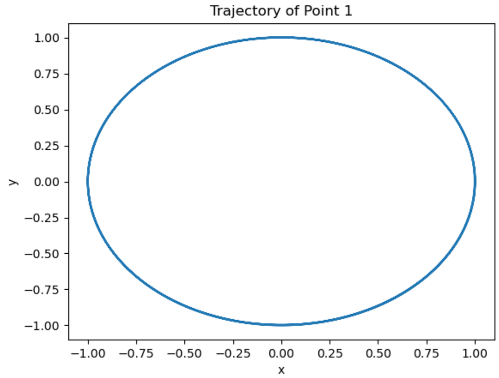

Figure 2. Visual representation of the orbital trajectory of a single surface point under the conditions $A = 1.74 \times 10^{-34}$, $B = 2.64 \times 10^{-5}$, and $C = 1.00 \times 10^{-5}$, resulting in an average radius of $r = 0.999765823920226$ and average angular velocity of $\omega = 0.3000660600495767$.

The circular trajectory and the preservation of radius and angular velocity in Figure 2 confirms that the moon maintains its original shape and radius over time in this case. This validates that our model can replicate the Moon's real-world stability when appropriately scaled.

Lateral Stiffness Dominates Structural Stability in the Model

We performed parameter sweeps for A, B, and C to examine how our system responded to different conditions. The value of r was calculated in each iteration after a fixed duration of $T = 100$, allowing us to quantify the relative impact of electrostatic, lateral spring, and radial spring forces on deformation.

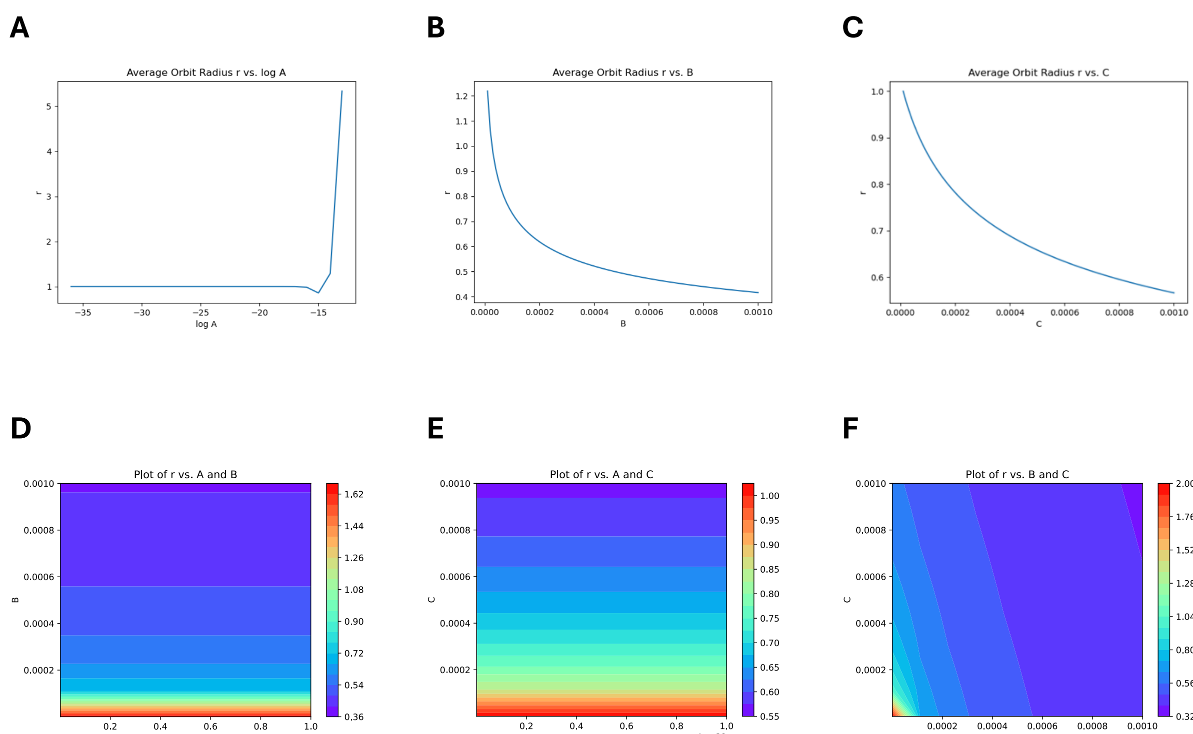

Figure 3. Quantifications of the effects of nondimensional force coefficients on average radius r. Panels **(A)-(C)** show the system's sensitivity to individual coefficients: **(A)** coefficient A (a logarithmic scale was used for the x-axis), **(B)** coefficient B, and **(C)** coefficient C. Additionally, contour plots were generated in panels **(D)-(F)** to examine the effects of pairs of parameters: **(D)** A and B, **(E)** A and C, and **(F)** B and C. Any coefficient not being varied was held constant at its real-world value as defined in the section above.

Figure 3 shows that the moon's structure responds differently according to each variable. According to panel (A), increasing A only results in a noticeably greater radius when its value is $\approx 10^{-15}$. As such, changes in the electrostatic strength, represented by A, exhibits virtually no effect on planet radius, as shown by panels (A), (D), and (E).

In contrast, increasing the other coefficients results in a smaller radius. As B increases, r drops significantly, indicating substantial deformation. The same result is shown for C, representing the relative radial stiffness, although with a more gradual slope, suggesting a comparatively small influence on planet shape. The slopes of the contour lines in panel (F) are $<<< -1$, confirming the system is the most sensitive to changes in B, which represents lateral stiffness.

Higher Stiffness Leads to Increased Average Angular Velocity

We performed additional parameter sweeps for A, B, and C. Average angular velocity (ω) was calculated to examine the effect of these coefficients on the planet's rotational dynamics.

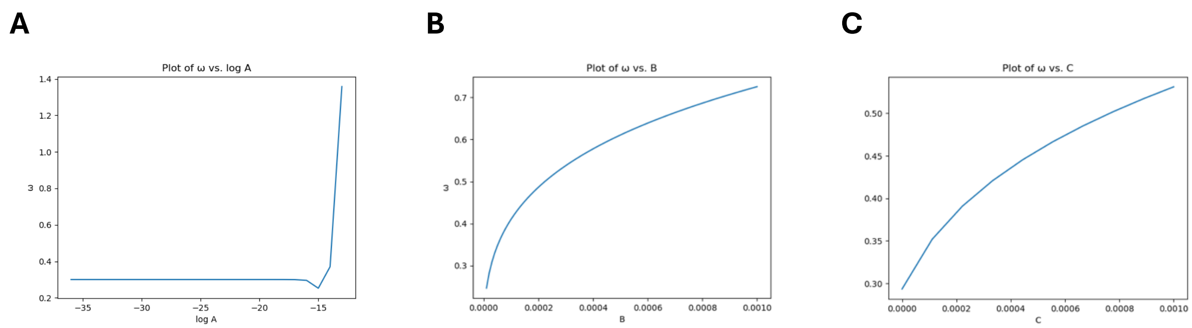

Figure 4. Quantifications of the effects of **(A)** A (a logarithmic scale was used for the x-axis), **(B)** B, and **(C)** C, on average angular velocity ω, calculated by the formula $\omega = \left|\vec{v} - \frac{\vec{v}\cdot\vec{r}}{\vec{r}\cdot\vec{r}}\vec{r}\right|$. As before, any coefficient not being varied was held constant at its real-world value.

As shown in Figure 4, ω is positively correlated with A, B, and C. As in the previous section, increasing A only influences ω when its value is $> \approx 10^{-15}$. On the other hand, increases in B and C produce significant and sustained growth in ω, with B producing a steeper rise. This suggests that lateral stiffness has the strongest effect on rotational dynamics, further emphasizing its dominant role in impacting planet deformation.

Convergence to Stable Elliptical Geometry Despite Explosive Behavior Through Self-Stabilization

A striking discovery of our model is its ability for self-stabilization. Notably, this phenomenon is observed even under extreme parameter configurations. Although the system may begin with significant deformation or "explosions" due to overly strong internal forces, it does not disintegrate or collapse. Instead, it rearranges itself and evolves towards a novel, stable configuration.

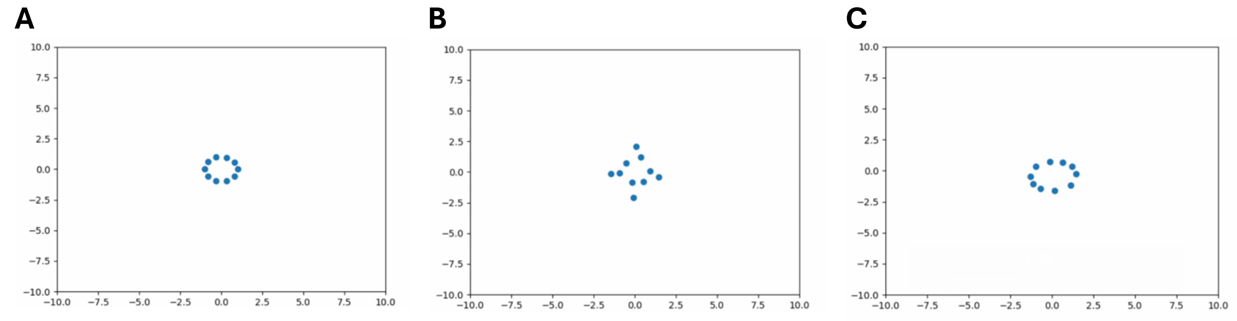

Figure 5. Images taken from three different stages during a simulation run, conducted with parameters $A = 1.00 \times 10^{-14}, B = 2.64 \times 10^{-5}$, and $C = 1.00 \times 10^{-5}$. **(A)** was taken during the initial state, **(B)** was taken during a moment of instability, and **(C)** was taken after self-stabilization, the final result a stable, elliptical structure.

Figure 5 depicts the evolution of the system from its initial state to its instable state to its self-stabilized state. The simulation was run with extreme conditions, namely A was magnified by a factor of order 10^{20} compared to its real-world value.

Conservation of Energy Proof (appendix?)

We conducted a detailed analytical derivation of the energy dynamics governing the system to confirm the physical validity of our model. In the following section, we prove that the total mechanical energy of our system is conserved under internal forces, namely electrostatic repulsions and spring forces. We also show that energy is not conserved when external forces (from the Earth and Sun) are introduced.

Electrostatic Forces

We begin by calculating the time derivatives of kinetic and potential energy. We use the equation

$$\frac{dK_i}{dt} = \vec{F_i} \cdot \vec{v_i}$$

where F_i is the net gravitational force and v_i is the velocity of an arbitrary point i. We know that

$$F_i = \sum_{j \neq i} \kappa \frac{q_i q_j}{r_{ij}^2},$$

where κ is the Coulomb constant and r_{ij} is the distance between the two distinct points i and j. We can write this as a vector

$$\vec{F_i} = -\sum_{j \neq i} \kappa \frac{q_i q_j}{r_{ij}^3} \cdot \vec{r_{ij}}$$

where $\vec{r_{ij}}$ is the vector from point j to i. Let (x_i, y_i) be the position of point i and (x_j, y_j) be the position of point j. We can rewrite our vector as

$$\vec{F_i} = -\sum_{j \neq i} \kappa \frac{q_i q_j}{r_{ij}^3} \cdot \begin{pmatrix} x_i - x_j \\ y_i - y_j \end{pmatrix}.$$

We can now calculate

$$\frac{dK_i}{dt} = \vec{F_i} \cdot \vec{v_i} = -\sum_{j \neq i} \left(\kappa \frac{q_i q_j}{r_{ij}^3} \cdot \begin{pmatrix} x_i - x_j \\ y_i - y_j \end{pmatrix} \right) \cdot \begin{pmatrix} dx_i/dt \\ dy_i/dt \end{pmatrix}.$$

Let us now calculate the rate of change of the gravitational potential energy of the system. We use the equation

$$\frac{dU_i}{dt} = \sum_{j\neq i} \frac{d}{dt}\left(-\kappa \frac{q_i q_j}{r_{ij}}\right) = \sum_{j\neq i} \kappa \frac{q_i q_j}{r_{ij}^2} \cdot \frac{dr_{ij}}{dt}.$$

By the chain rule,

$$\frac{dr_{ij}}{dt} = \frac{\partial r_{ij}}{\partial x_i}\frac{dx_i}{dt} + \frac{\partial r_{ij}}{\partial y_i}\frac{dy_i}{dt}.$$

We also know that $r_{ij} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}$. From here, we can find

$$\frac{\partial r_{ij}}{\partial x_i} = \frac{x_i - x_j}{r_{ij}}, \quad \frac{\partial r_{ij}}{\partial y_i} = \frac{y_i - y_j}{r_{ij}}.$$

Substituting, we get $\frac{dr_{ij}}{dt} = \frac{x_i - x_j}{r_{ij}}\frac{dx_i}{dt} + \frac{y_i - y_j}{r_{ij}}\frac{dy_i}{dt}$.

Writing this in vector notation, we get the following expression:

$$\frac{dr_{ij}}{dt} = \frac{1}{r_{ij}}\begin{pmatrix} x_i - x_j \\ y_i - y_j \end{pmatrix} \cdot \begin{pmatrix} dx_i/dt \\ dy_i/dt \end{pmatrix}.$$

Substituting this into our expression for $\frac{dU_i}{dt}$ we get

$$\frac{dU_i}{dt} = \sum_{j\neq i}\left(\kappa \frac{q_i q_j}{r_{ij}^3} \cdot \begin{pmatrix} x_i - x_j \\ y_i - y_j \end{pmatrix}\right) \cdot \begin{pmatrix} dx_i/dt \\ dy_i/dt \end{pmatrix} = -\frac{dK_i}{dt}.$$

We can clearly see that $\frac{dK_i}{dt} + \frac{dU_i}{dt} = 0$,

meaning that the total energy of the system is conserved under gravitational interactions.

Hooke's Law Energy, k_a

Hooke's Law tells us that

$$\vec{F_\imath} = -k_a \overrightarrow{r_{\imath,\imath+1}},$$

where k_a is the spring constant between the outer point masses.

The rate of change of kinetic energy is then

$$\frac{dK_i}{dt} = \vec{F_\imath} \cdot \vec{v_\imath} = -k_a \overrightarrow{r_{\imath,\imath+1}} \cdot \vec{v_\imath}.$$

The potential energy of a spring is defined by the equation

$$U_i = \frac{1}{2} k_a \overrightarrow{r_{\imath,\imath+1}}^2.$$

We differentiate to find the rate of change of potential energy:

$$\frac{dU_i}{dt} = k_a \overrightarrow{r_{\imath,\imath+1}} \cdot \frac{d\overrightarrow{r_{\imath,\imath+1}}}{dt} = k_a \overrightarrow{r_{\imath,\imath+1}} \cdot \vec{v_\imath} = -\frac{dK_i}{dt}.$$

Thus, we have shown that

$$\frac{dK_i}{dt} + \frac{dU_i}{dt} = 0,$$

meaning that the total energy of the system is conserved under k_a spring force interactions.

Hooke's Law Energy, k_c

We can use a similar process to prove the same statement for k_c spring forces.

We have

$$\vec{F_\iota} = -k_c \overrightarrow{r_{\iota,11}},$$

where k_c is the spring constant between the outer point masses and the origin.
Since the vector $\overrightarrow{r_{\iota,\iota+1}}$ has no particular computational significance in our proof for the k_a case, we can replace it with $\overrightarrow{r_{\iota,11}}$ and carry out the proof in the same manner we did previously to show that the total energy of the system is also conserved under k_c spring force interactions.

External Forces

The rate of change of kinetic energy is given by:

$$\frac{dK_i}{dt} = G\frac{m_i M_{12}}{r_{i,12}^3}\begin{pmatrix} x_i - x_{12} \\ y_i - y_{12} \end{pmatrix} \cdot \begin{pmatrix} dx_i/dt \\ dy_i/dt \end{pmatrix} + G\frac{m_i M_{13}}{r_{i,13}^3}\begin{pmatrix} x_i - x_{13} \\ y_i - y_{13} \end{pmatrix} \cdot \begin{pmatrix} dx_i/dt \\ dy_i/dt \end{pmatrix},$$

where (x_{12}, y_{12}) is the position of the Earth and (x_{13}, y_{13}) is the position of the Sun.
But unlike internal forces, there is no potential energy change due to external forces, so $\frac{dU}{dt} = 0$.
As such,

$$\frac{dK_i}{dt} + \frac{dU_i}{dt} = \frac{dK_i}{dt} \neq 0,$$

meaning that the total energy of the system is **not** conserved under external forces.

Net Energy Equation

Now, we combine the expressions for the change of kinetic energy and potential energy due to each of the forces to find the net energy equation:

$$\frac{dK_i}{dt} = \sum_{j\neq i}\left(\kappa\frac{q_i q_j}{r_{ij}^3}\cdot\overrightarrow{r_{\iota J}}\right) + \left(-k_a\overrightarrow{r_{\iota,\iota+1}} - k_a\overrightarrow{r_{\iota,\iota-1}} - k_c\overrightarrow{r_{\iota,11}} - G\frac{m_i m_{12}}{r_{i,12}^3}\cdot\overrightarrow{r_{\iota,12}} - G\frac{m_i m_{13}}{r_{i,13}^3}\cdot\overrightarrow{r_{\iota,13}}\right)\cdot\frac{d\vec{x_\iota}}{dt}$$

$$\frac{dU_i}{dt} = -\sum_{j\neq i}\left(\kappa\frac{q_i q_j}{r_{ij}^3}\cdot\overrightarrow{r_{\iota J}}\right) + \left(-k_a\overrightarrow{r_{\iota,\iota+1}} - k_a\overrightarrow{r_{\iota,\iota-1}} - k_c\overrightarrow{r_{\iota,11}}\right)\cdot\frac{d\vec{x_\iota}}{dt}$$

Adding these two equations gives us an expression for the change in mechanical energy for body i:

$$\frac{dE_i}{dt} = \frac{dK_i}{dt} + \frac{dU_i}{dt} = \left(-G\frac{m_i m_{12}}{r_{i,12}^3}\cdot\overrightarrow{r_{\iota,12}} - G\frac{m_i m_{13}}{r_{i,13}^3}\cdot\overrightarrow{r_{\iota,13}}\cdot\right)\cdot\frac{d\vec{x_\iota}}{dt}$$

We can now calculate the total change in mechanical energy of our system by summing throughout the N bodies:

$$\frac{dE}{dt} = \sum_{i}^{N}\left(-G\frac{m_i m_{12}}{r_{i,12}^3}\cdot\overrightarrow{r_{\iota,12}} - G\frac{m_i m_{13}}{r_{i,13}^3}\cdot\overrightarrow{r_{\iota,12}}\right)\cdot\frac{d\vec{x_\iota}}{dt} \neq 0.$$

Thus, we have found that only external forces are capable of causing a change in the mechanical energy of the system.

[This part may omit]
Second term on the right side of above expression is the rate of change of kinetic energy due to external forces whereas first term on the right side is rate of change of kinetic energy due to forces between bodies in our system or internal forces.

$$U_i = \sum_{j \neq i} \left(-G \frac{m_i m_j}{r_{ij}} + \kappa \frac{q_i q_j}{r_{ij}} \right) + \frac{1}{2} k r_{i,i-1}^2 + \frac{1}{2} k r_{i,i+1}^2$$
$$- \sum_{j \neq i} G \frac{m_j}{r_{ij}^3} \cdot \overrightarrow{r_{ij}} + \sum_{j \neq i} \kappa \frac{q_i q_j}{r_{ij}^3 m_i} \cdot \overrightarrow{r_{ij}} - \sum_i \frac{k_a \overrightarrow{r_{i,i+1}}}{m_i} - \sum_i \frac{k_c \overrightarrow{r_{i,11}}}{m_i} - d\left(v_i \cdot \widehat{r_{i,11}}\right) \widehat{r_{i,11}}.$$

???? Include damping in energy proof?

V. Discussion

Insignificance of Electrostatic Forces Under Realistic Parameters
Our results show that the internal electrostatic forces of the Moon, represented by A, minimally contribute to deformation. In fact, significant changes of r are only observed when the value of A exceeds $\approx 10^{-15}$. Comparing this to the real-world value ($A = 1.74 \times 10^{-34}$), this indicates that the Moon must have nearly 10^{19} times its current charge to significantly impact its structure, implying that under realistic astrophysical conditions, the Moon's net charge has an indiscernible effect. Resultingly, electrostatic forces can be safely neglected in our Moon's deformation models unless extreme charge accumulation occurs.

Unexpected Lateral Spring Force Sensitivity
Contrary to our initial expectations, coefficient B was found to exert a stronger influence on the Moon's structure than C. Intuitively, one might assume that the radial spring forces would dominate. Its directions directly pull towards the center, as opposed to the directions of the lateral spring forces, which are tangential and should not have much effect. However, our results suggest that lateral spring forces are dominant. This phenomenon may be explained by our limited discretization of the model (with $N = 10$ point masses), which amplifies the influences of the lateral spring forces. In future studies, increasing N or transitioning into a continuous elastic shell may produce more realistic radial dominance.

Comparison of Radius and Angular Velocity
A striking pattern emerges when we compare the graphs of the average radius and angular velocity over time. Figures 3A and 4A showcase nearly identical trends, suggesting a strong, positive correlation between r and ω, specific to electrostatic forces. In contrast, Figures 3B and 4B, as well as Figures 3C and 4C, exhibit a negative correlation between r and ω. This result implies that both repulsive and attractive forces contribute to increased angular motion, which may act as a stabilizing mechanism, possibly redistributing internal forces to preserve structural equilibrium.

Implications of the Self-Stabilizing Effect

Our system, as described above, consistently evolved towards a stable elliptical configuration, despite any explosions or severe deformation. Internal damping dissipates excess energy. Combined with the effect of elastic forces, the body settles into a new equilibrium state. This self-stabilization implies that a system influenced by gravitational, electrostatic, elastic forces, and internal damping can recover from extreme structural disturbances without external intervention. It also suggests that the system is following a path toward a minimum-energy configuration, governed by a balance between internal forces and damping. Analytical work using energy-based methods, such as Lyapunov stability analysis or energy landscape modeling, may be conducted to predict and characterize deformation.

Future Studies [change title of section]
Future research can expand on our model in several keyways. Our model represents the Moon as having only one layer of mass, but the physical structure is not hollow. It also assumes the Moon consists of point masses instead of a continuum system, which may be limited in determining all the physical factors affecting planetary deformation. However, the simplicity of our model allows us to understand the fundamentals behind planetary deformation without being overly computational.

One potential avenue is to incorporate multiple layers of mass to better simulate the characteristics of real celestial bodies. Another promising direction is to develop our model into a three-dimensional simulation that accounts for depth in the Z-axis. Additionally, celestial bodies are rarely perfect spheres, or their shapes could have already been severely deformed due to tidal forces. Rather than approximating bodies as perfect circles, future work can be done to model them as triaxial ellipsoids, a model used in other notable studies (Akinsanmi et al., 2019, p. 2). Finally, incorporating the gravitational influence of a third external body, namely Jupiter, could provide a more precise model of lunar deformation. Adding these features would enhance the model's realism and increase its applicability to a wider range of astrophysical scenarios.

VI. Conclusion

Our simulations indicate that electrostatic forces have minimal impact in lunar deformation under physically realistic parameters. They also reveal that lateral spring forces dominate over radial, gravitational, and electrostatic forces in inducing structural changes, specifically in radii. Finally, our model’s ability to self-stabilize regardless of extreme perturbations demonstrates that the Moon’s structure has a high level of resilience.

The modeling framework we developed in this study is not limited to the Moon. It becomes a versatile tool for probing the deformation of any planetary body. Through our successful development of a model that captures the interplay between elasticity, gravity, and electrostatics, this work advances us toward a physics-based theory of planetary structural stability across the Solar System.